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Renal protective effect and action mechanism of Huangkui capsule and its main five flavonoids

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quercetin-3′-*O*-*β*-D-glucoside (PubChem CID: 44259229)
isoquercitrin (PubChem CID: 5280804)
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ABSTRACT

Ethnopharmacological relevance: The flower of *Abelmoschus manihot* (Linn.) Medicus (*A.* manihot), as a traditional Chinese Herbal medicine, was used widely in China with efficacy of inducing diuresis for treating strangurtia, and subdhing swelling and detoxicating. It has been reported that Huangkui capsule, prepared by the extract of the flower of *A. manihot*, can reduce the content of urinary protein, serum creatinine and serum urea nitrogen in nephropathy rats and processes renoprotective activity, while the action mechanism need to illuminate deeply.

Aims of the study: In this study, we investigated the protection effect of Huangkui capsule on tubulointerstitial fibrosis in chronic renal failure (CRF) rats and its mechanism against high glucose-induced epithelial to mesenchymal transition (EMT) in renal tubular epithelial cells (HK-2) of its bioactive components.

Materials and methods: The animals were divided into normal group, CRF model group and Huangkui capsule-treated group. Hematoxylin eosin (HE) staining and Masson staining were applied to observe pathological changes in renal tissue of different groups. Biochemical indicators including serum urea nitrogen (BUN), urine protein (UP) and serum creatinine (Scr) were measured according to the manufacturer's instructions of kits. HK-2 cell damaged model was established to access the protection effect and action mechanism of five main flavonoids from Huangkui capsule. The experimental cells were divided into eight groups: control group, model group, positive drug group and five main flavonoids treated groups. The dichlorodihydrofluorescein diacetate (DCFH-DA) assay was used to determine the reactive oxygen species (ROS) in different groups. Western blot was applied to analyze the expression of pathogenesis-related proteins in different groups.

Results: The results stated that Huangkui capsule significantly inhibited the elevation of Scr, BUN, UP, the expression of α-smooth muscle actin (α-SMA), phosphorylation-extracellular signal-regulated kinase (p-ERK1/2), NADPH Oxidase 1, NADPH Oxidase 2 and NADPH Oxidase 4 in adenine-induced CRF rats. The main bioactive components of quercetin (QT), hyperoside (HY), isoquercitrin (IQT), gossypetin-8-*O*-*β*-D-glucuronide (GG) and quercetin-3′-*O*-glucoside (QG) at the dosage of 100 μM, like NADPH oxidase inhibitor diphenyleneiodonium, exhibited a significant effect on inhibiting the expression of α-SMA, p-ERK1/2, NADPH Oxidase 1, NADPH Oxidase 2 and NADPH Oxidase 4 in high glucose-induced HK-2 cells, especially GG.

Abbreviation: CRF, Chronic renal failure; ESRD, end-stage renal disease; EMT, Epithelial to mesenchymal transition; HK-2, Human kidney proximal tubule epithelial cell; QT, quercetin; IQT, isoquercitrin; HY, hyperoside; GG, gossypetin-8-*O*-*β*-D-glucuronide; QG, quercetin-3′-*O*-glucoside; NOX1, NADPH Oxidase 1; NOX2, NADPH Oxidase 2; NOX4, NADPH Oxidase 4; α-SMA, α-smooth muscle actin; p-ERK1/2, phosphorylation-extracellular signal-regulated kinase

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Conclusions: These results demonstrated that Huangkui capsule and the flavonoids components prevent tubulointerstitial fibrosis in CRF rat involvement in the action mechanism of inhibiting NADPH oxidase/ROS/ERK pathway.

1. Introduction

Chronic renal failure (CRF) is a chronic disease of renal insufficiency. It is characterized by metabolites retention, electrolyte balance disorder and renal endocrine dysfunction in clinic. The crucial manifestations of the disease included tubulointerstitial fibrosis, low estimated glomerular filtration rate and albuminuria and so on. Tubulointerstitial fibrosis is the most common induction factor of end-stage renal disease (ESRD), the last stage of CRF (Afkarian et al., 2013; Liu, 2011). Epithelial to mesenchymal transition (EMT) in human renal tubular epithelial (HK-2) cells played an important role in the initiation of tubulointerstitial fibrosis and was involved in the reversible progression (Carew et al., 2012; Liu, 2006). As a result, inhibiting EMT is a key way to prevent the process of tubulointerstitial fibrosis. It has been reported that high glucose had the potential ability to induce EMT in HK-2 cells (Lee et al., 2013; Wang et al., 2014; Wei et al., 2013). In addition, high glucose has been shown to be able to promote the generation of intracellular reactive oxygen species (ROS), mainly derived from nicotinamide adenine dinucleotide phosphate (NADPH) oxidases and playing a key role in EMT and tubulointerstitial fibrosis (Chen et al., 2012; Ha et al., 2005; Rhyu et al., 2005). Accordingly, it is an appropriate approach to neutralize the excessive ROS in treating with high glucose-induced EMT and renal fibrosis.

Traditional Chinese medicine (TCM) with multi-components and multi-targets has been used to treat CRF in China and has obtained its popularity due to its efficacy and low side-effects (Ye, 2004). The flower of *Abelmoschus manihot* (Linn.) Medicus (*A. manihot*), as a traditional Chinese Herbal medicine, was used widely in China with efficacy of inducing diuresis for treating strangurtia, and subdhing swelling and detoxicating. It has been used to treat chronic glomerulonephritis for many years in China (Jiang et al., 2012; Shen et al., 2014). Huangkui capsule, prepared by the extract of flower of *A. manihot*, gained regulatory approval from China's State Food and Drug Administration for the treatment of chronic nephritis in 1999 (Want et al., 2005). The bioactive components mainly in Huangkui capsule contained quercetin (QT), isoquercitrin (IQT), hyperoside (HY), gossypetin-8-*O*-*β*-D-glucuronide (GG) and quercetin-3′-*O*-glucoside (QG) (Liu et al., 2011). It was reported that QT ameliorated early diabetic renal hyperdynamic abnormality and renal tubular EMT and renal fibrosis in streptozotocin-induced diabetic rats (Guo et al., 2010; Xu et al., 2001; Lai et al., 2012, 2015). It is necessary to evaluate the components′ therapeutical effects on renal fibrosis and in further to illuminate the action mechanism.

The aim was to study the therapeutical effect of Huangkui capsule on tubulointerstitial fibrosis in adenine-induced CRF rats. And the other aim is to evaluate the effect of active components on inhibiting EMT in Human kidney proximal tubule epithelial (HK-2) cells in further to explore the action mechanism.

2. Materials and methods

2.1. Chemicals and instruments

Huangkui capsule (16060216) was obtained from SZYY Group Pharmaceutical Limited (Jiangyan, China). The contents of QT, HY, IQT, G-8-G and Q-3-G have been determined by our pervious study, which is 0.251%, 1.21%, 0.91%, 0.705% and 1.77%, respectively by UPLC method (Lu et al., 2013). And the UPLC Chromatogram map of Huangkui capsule and MS/MS of five flavonoids has been acquired (Cai et al., submitted for publication, Data in Brief). QT, HY, IQT, G-8-G and Q-3-G were prepared from *Abelmoschus moschatus* (L.) medic by ourselves and the purity was above 95%; Adenine (S18009, 98%) was bought from shanghai inke biological technology co., LTD (Shanghai, China). Serum urea nitrogen (BUN) reagent kit, serum creatinine (Scr) reagent kit, urine protein (UP) reagent kit were bought from Nanjing Jiancheng Bioengineering Institute (Nanjing, China); HK2 cells, penicillin/streptomycin solution (KGY002), 0.25% Tripsin-EDTA (KGY001), PBS (KGB500) and reactive oxygen species assay kit (KGT010-1) were purchased from Nanjing keygen technology development co., LTD. Cell culture flask (USA, FALCON, 353014); DMEM (USA, GIBCO, 12800-082); F12 (USA, GIBCO, 883684); FBS (USA, ExCell Biology, FBS500); The primary antibodies against NOX1, NOX2, NOX4, α-SMA, E-cadherin, fibronectin, and β-actin were obtained from Abcam (Cambridge, UK). Anti-pERK1/2 (Thr202/Tyr204) and anti-ERK1/2 antibodies were from Cell Signaling Technology (Beverly, MA, USA). Diphenyleneiodonium (DI, NADPH oxidase inhibitor) were purchased from Calbiochem Corp. (La Jolla, CA).

Clean Bench (China, Suzhou sinmo purification technology co., LTD, SW-CJ-1FD); 6 well cell culture plate (USA, Corning Incorporated, 3516); CO_2 incubators (Japan, SANYO, XD-101); inverted biological microscope (Japan, OLYMPUS, IX51); table-top low speed centrifuge (Shanghai Medical Instruments (Group) Ltd., corp., 80-2); Flow cytometry (USA, Becton-Dickinson, FACS Calibur).

2.2. Animals

Male SD rats were obtained from the Central Animal Breeding House of Nanjing University of Chinese Medicine (Nanjing, China). They were housed in cages with a constant humidity (ca. 60% ± 2%) and temperature (ca. 25 ± 1 °C) and with a light/dark cycle of 12 h. The animals were used when 6 weeks and underwent an adaptation period of several days, during which they were free access to water and food.

The 18 rats weighing 190–210 g were randomly divided into control groups, model group and Huangkui capsule group (N = 6/group). According to the modeling method of CRF in the reference (Hu et al., 2011), model group and Huangkui capsule group were given 150 mg/kg·d body weight of adenine dissolved with 1% (w/v) gum acacia solution (by gastric gavage) for 28 days. From the 15th day to the 28th day, Huangkui capsule group was simultaneously given 0.75 g/kg·d body weight of Huangkui capsule content. Control group received physiological saline of equal volumes for 28 days. Body weight was measured for all rats every two days. All experimental procedures were carried out in accordance with the Guide for the Care and Use of Laboratory Animals. And before the animal experiments were carried out, the procedures were approved by the Research Ethical Committee of Nanjing University of Chinese Medicine (No.1607260004, Nanjing, China).

2.3. Sample collection

We did our best in order to minimize the suffering of experimental animals and to use the number of rats which enables us to generate reliable data. After 14 days' administration cycle, rats with water but no food for twelve hours were anesthetized with ethyl ether, and blood samples were obtained by abdominal aorta, each urine sample was separately obtained via metabolic cages in 24 h. The left and right renal tissues were harvested and immediately washed with physiological saline. The left one was stored at 10% formalin solution for subsequent pathological and the right one was stored at −80 °C for western blot. The blood and urine samples were stored at −80 °C for the determination of biochemical indicators.

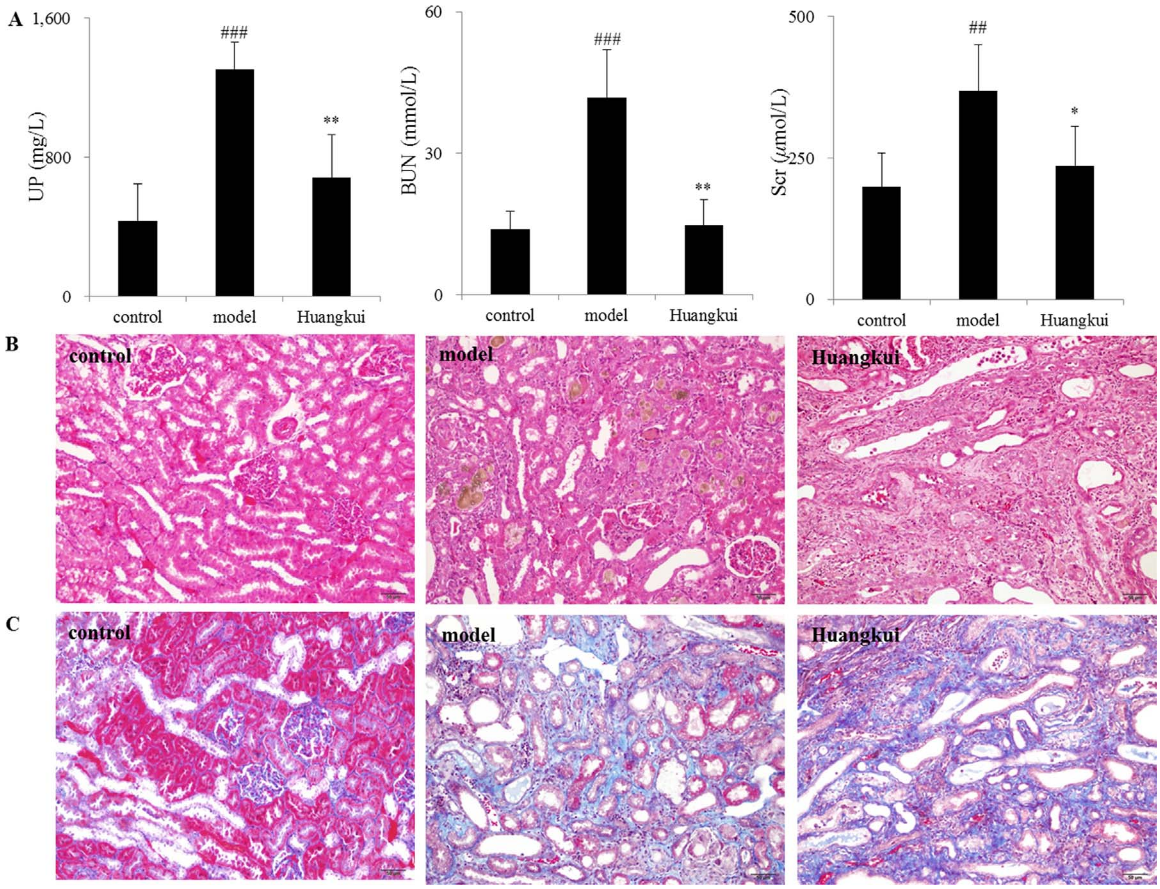

Fig. 1. Determination of biochemical indicators and analysis of renal pathological section among control group, model group and Huangkui capsule-administered group. (A) Determination of Serum urea nitrogen (BUN), Serum creatinine (Scr) and Urine protein (UP). (B) HE staining of kidney biopsy (×400). (C) Masson staining of kidney biopsy (×400). ##$p < 0.01$; ###$p < 0.001$: CRF models vs controls; *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$: after treatment by Huangkui capsule vs CRF models.

2.4. Biochemical indicators measurements and pathological analysis

The therapeutic efficacy on CRF was evaluated for the levels of BUN and UP in urine and Scr in plasma, which was detected followed by the description supplied by kits manufacturer. Part of the renal tissues samples were performed for hematoxylin eosin (HE) staining and Masson staining in order to observe pathological changes in renal tissue, degrees of fibrosis tissue hyperplasia, structures of glomeruli and tubules, arrangement of tubules and urate deposition through electron microscope (×400).

2.5. Cell culture

HK-2 cells were cultured in 90% DMEM/F12 containing 5 mM glucose supplemented with 10% FBS in an incubator keeping the conditions of 37 °C, 5% CO_2 and saturated humidity. The cells were sub-cultured in 6-well plates at a density of 1×10^6 cell/well 24 h prior to experiments.

2.6. Treatment with high-glucose and flavonoids compounds isolated from flower of A. moschatus

The HK2 cells were divided into eight groups. The normal group (NG) was cultured with 5 mM glucose. The model group (HG) was treated with high glucose (25 mM) for 24 h. The positive control group (DI) was cultured with high glucose (25 mM) and diphenylene iodonium (10 μM). The quercetin-treated groups (QT) were processed with high glucose (25 mM) and quercetin (5, 10, 25, 50, 100 μM) for

Table 1
Pathological evaluation of kidneys from control group model group and Huangkui capsule-treated group rats (n = 6).

Compartment	Parameter	Average score[1]		
		Control group	Model group	Huangkui group
Glomeruli	atrophy	0	$1.00^{\#\#\#}$	$0.60 \pm 0.39^{*}$
Tubule	Tubular atrophy	0.40 ± 0.22	$2.40 \pm 0.45^{\#\#\#}$	$1.40 \pm 0.30^{**}$
	urate deposition	0	$3.00^{\#\#\#}$	$1.80 \pm 0.50^{***}$
	Morphology	0	$1.00^{\#\#\#}$	0.75^{***}
Interstitium	Inflammation	0	$2.00^{\#\#\#}$	$1.50 \pm 0.50^{*}$
	Fibrosis	0	$1.40 \pm 0.55^{\#\#\#}$	$1.00 \pm 0.58^{*}$

Data is presented as averagescore of six rats. #$p < 0.05$; ##$p < 0.01$: CRF models vs controls; *$p < 0.05$;**$p < 0.01$;***$p < 0.001$: after treatment by Huangkui capsule vs CRF models.

[1] the parameters glomeruli atrophy, tubular atrophy, tubular morphology, focal calcium deposits and interstitial fibrosis and interstitial inflammation are graded as follows: 0; affecting 0–5% of the renal area, 0.5; 6–25%, 1; 26–50%, 2; 50–75%, and 3; > 75%.

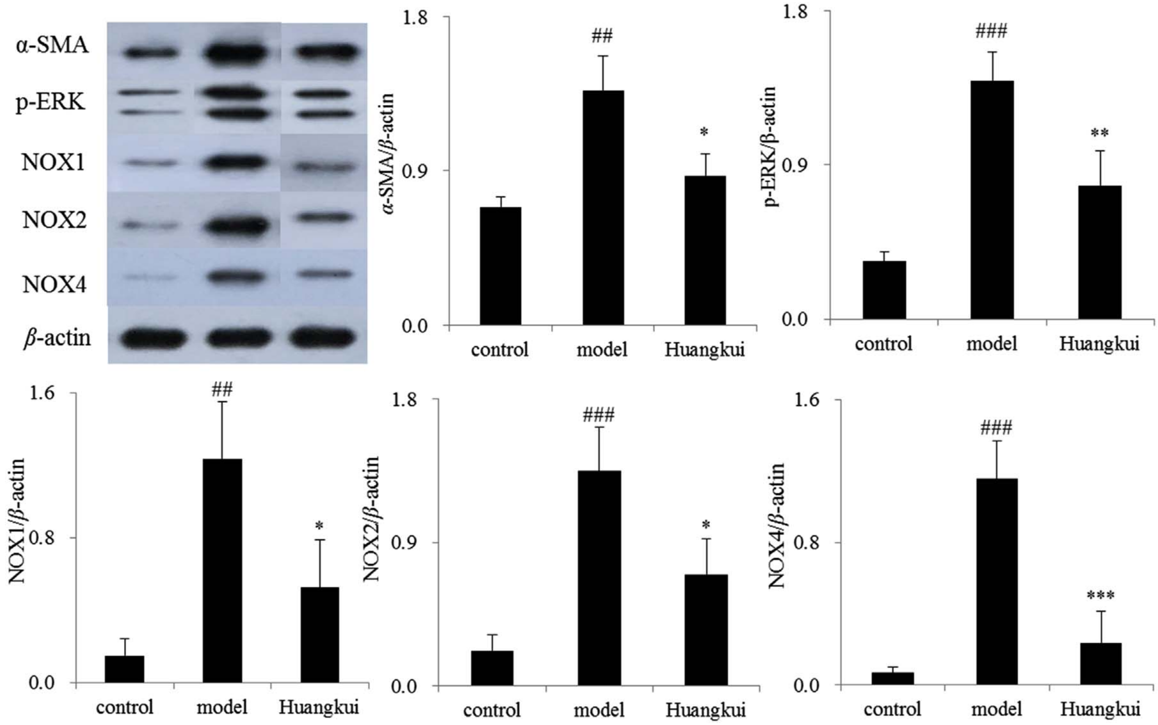

Fig. 2. The expression of NOX1, NOX2, NOX4, α-SMA, p-ERK in renal tissue of control rats, CRF rats, and Huangkui capsules-treated rats. ##p < 0.01; ###p < 0.001: CRF models vs controls; *p < 0.05;**p < 0.01;***p < 0.001: after treatment by Huangkui capsule vs CRF models.

24 h. The isoquercitrin-treated groups (IQT) were handled with high glucose (25 mM) and isoquercitrin (5, 10, 25, 50, 100 μM) for 24 h. The quercetin-3′-O-β-D-glucoside-treated groups (QG) were disposed with high glucose (25 mM) and quercetin-3′-O-β-D-glucoside (5, 10, 25, 50, 100 μM) for 24 h. The gossypetin-8-*O*-β-D-glucuronide-treated groups (GG) were conducted with high glucose (25 mM) and gossypetin-8-O-β-D-glucuronide (5, 10, 25, 50, 100 μM) for 24 h. The hyperoside-treated groups (HY) were dealed with high glucose (25 mM) and hyperoside (5, 10, 25, 50, 100 μM) for 24 h. In the set of experiments, these cells were prepared for Morphology observation, for the dichlorodihydrofluorescein diacetate (DCFH-DA) assay to determine the reactive oxygen species (ROS) and for western blotting to examine the protein expression of E-cadherin, α-smooth muscle actin (α-SMA), ERK, p-ERK, fibronectin, β-actin, NOX1, NOX2 and NOX4.

2.7. Measurement of intracellular ROS level

Intracellular ROS was detected using the Oxidation-sensitive 2′, 7′

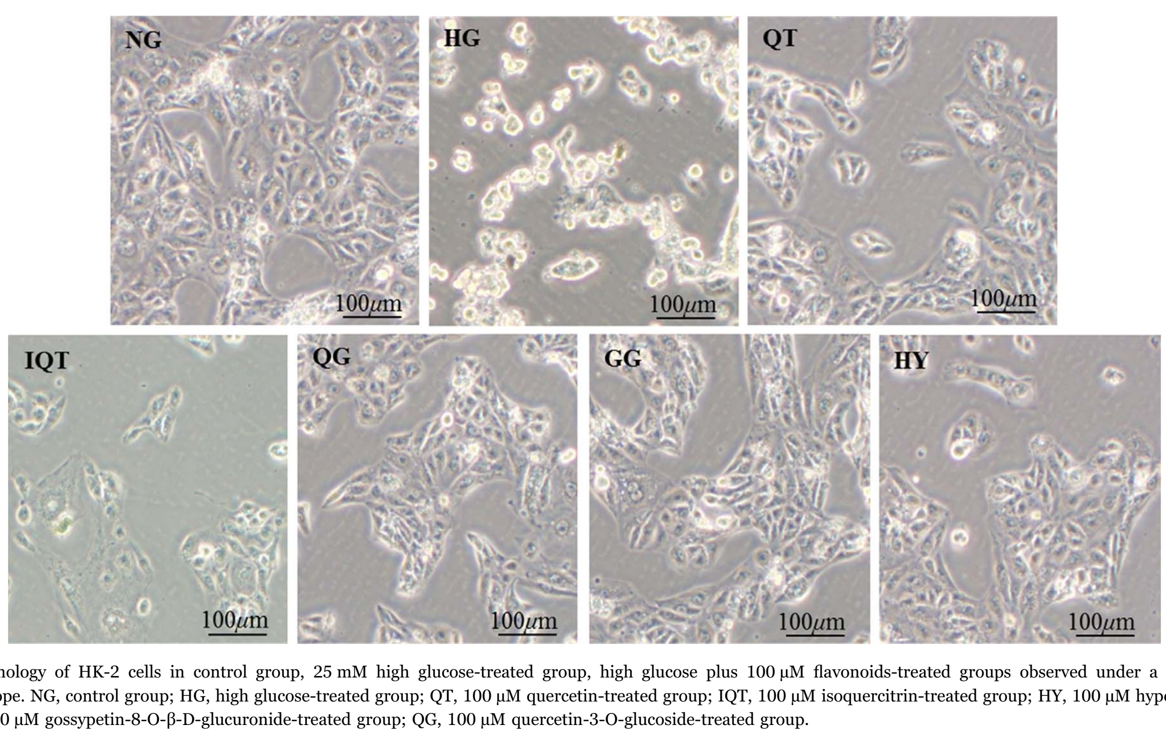

Fig. 3. Morphology of HK-2 cells in control group, 25 mM high glucose-treated group, high glucose plus 100 μM flavonoids-treated groups observed under a phase-contrast photomicroscope. NG, control group; HG, high glucose-treated group; QT, 100 μM quercetin-treated group; IQT, 100 μM isoquercitrin-treated group; HY, 100 μM hyperoside-treated group; GG, 100 μM gossypetin-8-O-β-D-glucuronide-treated group; QG, 100 μM quercetin-3-O-glucoside-treated group.

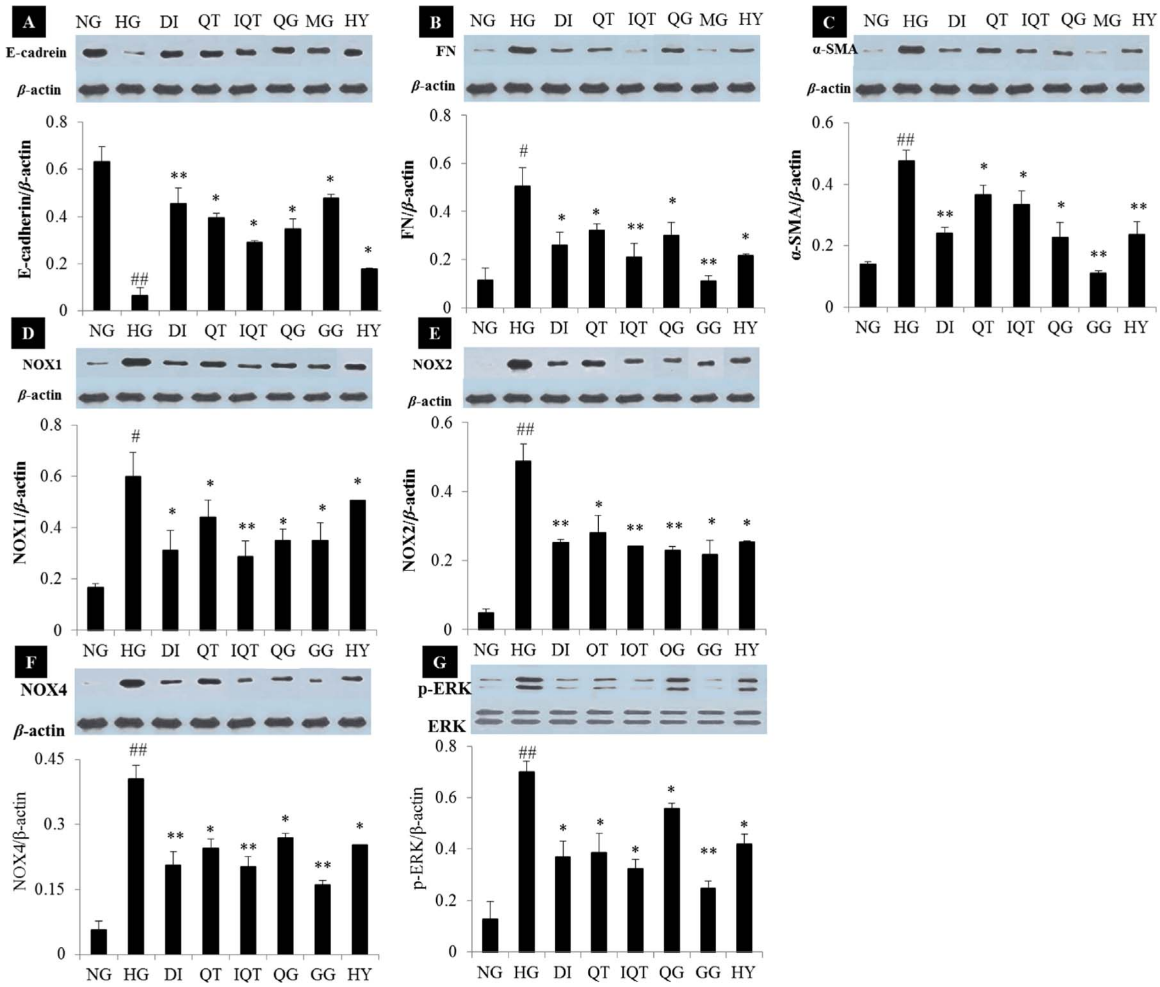

Fig. 4. The expression of NOX1, NOX2, NOX4, α-SMA, E-cadherin, fibronectin, and p-ERK in HK-2 cells. ##p < 0.01; ###p < 0.001: high glucose-treated group vs control group; *p < 0.05; **p < 0.01; ***p < 0.001: after treatment by Huangkui capsule vs high glucose- treated group. NG, control group; HG, high glucose-treated group; QT, 100 μM quercetin-treated group; DI, 10 μM positive drugs-treated group; IQT, 100 μM isoquercitrin-treated group; HY, 100 μM hyperoside-treated group; GG, 100 μM gossypetin-8-O-β-D-glucuronide-treated group; QG, 100 μM quercetin-3-O-glucoside-treated group.

-dichlorofluorescein diacetate (DCFH-DA) assay. Each group cells were cultured in 6-wellcell plates, making the cellular coverage reaches to 80~90%. These cells were washed once time with PBS and adjusting the cell concentrations to 1×10^6 /ml and then treated with 10 μmol/L DCFH-DA in serum-free DMEM for 20 min at 37 °C, shaken slightly every 3–5 min to keep a good contact with probe After washing thrice with serum-free cell culture medium to remove the residual DCFH-DA, in order to quantify the ROS production of these cells, a flow cytometry was used to detect DCF fluorescence at excitation and emission wavelengths of 488 nm and 525 nm, respectively.

2.8. Western blotting

Total protein extract from HK2 cells was resolved on 8% SDS–polyacrylamide gels and transferred to polyvinylidene fluoride (PVDF) membranes by a semi-dry transfer blotting system. After blocking for 2 h at 4 °C in blocking buffer (10% fat-free milk TBS with 0.1%Tween 20), the membranes were incubated overnight with primary antibodies, including α-SMA, E-cadherin, fibronectin, NOX1, NOX2, NOX4, p-ERK1/2, total ERK1/2, and β-actin. After horseradish peroxidase-labeled secondary antibodies were added at 37 °C for 1 h, the blots were visualized with the ECL-plus detection system. Analysis of gray was performed using Quantity One Software.

2.9. Statistical analysis

SPSS 16.0 software (SPSS Inc., Chicago, IL, USA) was applied for statistical analysis. Statistical results were expressed as mean ± standard deviation (SD). Comparisons between groups were made using one-way ANOVA followed by Tukey multiple comparison test. A p-value < 0.05 was regarded as statistically significant.

3. Results

3.1. Biochemical indicators measurements and pathological analysis

The analytical results of biochemical indicators were presented in Fig. 1A. After 28 days of establishing the model, Scr, BUN and UP in CRF rats increased significantly compared with control group ($p < 0.05$ or $p < 0.01$ or $p < 0.001$). After treatment with Huangkui capsule at 0.75 g/kg·d, the levels of these indicators were nearly restored to normal levels. The results of the histological examination were consistent with the biochemical analysis (see Fig. 1B and C). Compared with the healthy rats, renal tissues of the CRF rats exhibited glomerular and tubular atrophy, urate deposition, interstitial fibrosis and interstitial inflammation. The symptoms were obviously improved after treatment with Huangkui capsule. The pathological score in kidney is provided in Table 1.

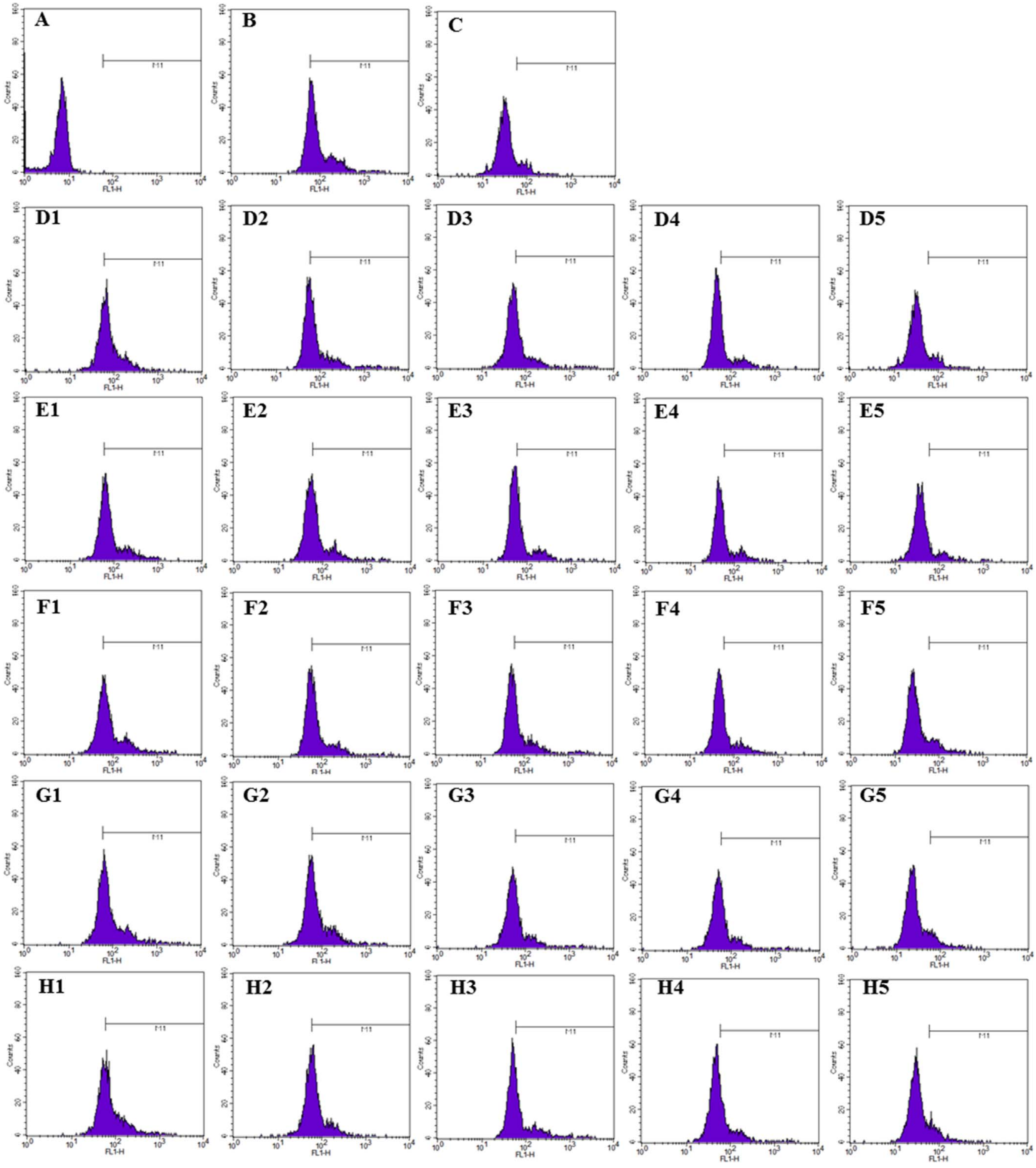

Fig. 5. Fluorescence intensity in HK-2 cells dealed with different drugs detected by flow cytometry. (A), 5 mM glucose; (B), 25 mM glucose; (C), 25 mM glucose plus 10 μM DI; (D), 25 mM glucose plus quercetin; (E), 25 mM glucose plus isoquercitrin; (F), 25 mM glucose plus quercetin-3-O-glucoside; (G), 25 mM glucose plus gossypetin-8-O-β-D-glucuronide; (H), 25 mM glucose plus hyperoside; Lable 1–5 stand for different dosages of 5, 10 ,25, 50, 100 μM in turn.

3.2. The expression of NOX1, NOX2, NOX4, α-SMA, p-ERK in renal tissue

As shown in Fig. 2, compared with control group, the expression of NOX1, NOX2, NOX4, α-SMA, p-ERK in renal tissue of CRF rats were significantly increased. After intervening in Huangkui capsule, the expressions of these proteins were decreased to the normal-like level.

3.3. High glucose-induced EMT in HK-2 cells and the effects of five flavonoids

To access the effects of high glucose and flavonoids on EMT in HK-2 cells, morphology of HK-2 cells among control group, model group, and five flavonoids-treated groups were observed under a phase-contrast photomicroscope. As shown in Fig. 3, HK-2 cells in control group

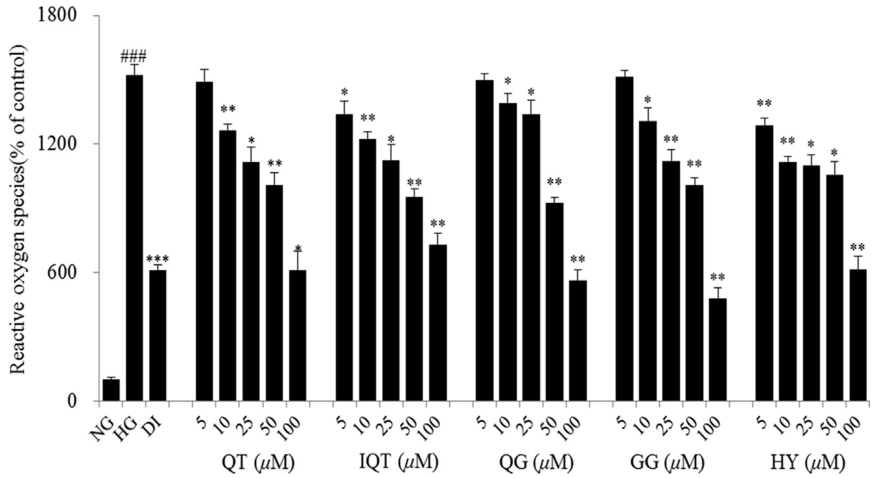

Fig. 6. Relative reactive oxygen soecies in HK-2 cells dealed with different drugs detected by flow cytometry. Values are expressed as mean ± SD. ##p < 0.01: High glucose vs normal glucose; *p < 0.05;**p < 0.01: after treatment by flavone morphons vs High glucose. NG, control group; HG, high glucose-treated group; QT, quercetin-treated group; DI, 10 μM positive drugs-treated group; IQT, isoquercitrin-treated group; HY, hyperoside-treated group; GG, gossypetin-8-O-β-D-glucuronide-treated group; QG, quercetin-3-O-glucoside-treated group.

maintained their cobblestone-like epithelial façade, and cells in model group processed the spindle-like fibroblastic shape. After treating with five flavonoids compunds, the appearances improved respectively.

α-SMA, a remarkable marker of myofibroblast and the critical marker of EMT (Wang et al., 2006), and fibronectin, a pivotal extracellular matrix protein, and E-cadherin, a typical phenotypic marker of epithelial cell were detected to assess the effect of high glucose on the induction of EMT in HK-2 cells by western blot analysis. As showed in Fig. 4A, B, C, high glucose significantly increased the expressions of α-SMA, fibronectin, but decreased the expression of E-cadherin, indicating high glucose had the property to induce EMT in HK-2 cells. QT, HY, IQT, QG and GG, regulating the expressions of *α*-SMA, fibronectin and E-cadherin, exhibited the ability to treat EMT in some extent. Noticeably, after treated by GG, the levels of *α*-SMA, fibronectin and E-cadherin were the closest to the normal level, presaging GG may process the strongest effect on curing high glucose-induced EMT in HK-2 cells.

3.4. High glucose increasing ROS production and the expressions of NADPH oxidase subunits NOX1, NOX2 and NOX4 in HK-2 cells and the inhibition effects of flavonoids compounds

Intracellular ROS plays a critical role in the induction of EMT and its generation process was associated with the expressions of NADPH oxidase subunits NOX1, NOX2 and NOX4 in HK-2 cells (Ha et al., 2005; Li et al., 2003). Hence, it was necessary to detect ROS levels and the expressions of NADPH oxidase subunits NOX1, NOX2 and NOX4 in HK-2 cells before and after high glucose-treated and flavonoids-treated. As emerged in Figs. 5 and 6, ROS production was observably increased in high glucose-induced HK-2 cells. After incubation with QT, HY, IQT, QG and GG at five different dosages, it was showed that the debasement of ROS was positive associated with the rise of dosages in some degree. The results from western blot analysis demonstrated a significant increase in the expressions of NOX1, NOX2 and NOX4 in HK-2 cells after high glucose-induced, and a remarkable decrease via treating with 100 μM QT, HY, IQT, QG and GG (Fig. 4D, E, F).

3.5. High glucose activating ERK1/2 in HK-2 cells and flavonoids compounds blocking its activation

ERK1/2 is a critical cellular signaling molecule associated with EMT occurring in different organs (Rhyu et al., 2005; Zeisberg et al., 2001). Herein, the expression of *p*-ERK1/2 was investigated by western blotting analysis. As the results shown in Fig. 4G, high glucose significantly induced the phosphorylation of ERK1/2, suggesting that ERK1/2 signaling may be involved in high glucose-induced EMT in HK-2 cells. After incubation with 100 μM QT, HY, IQT, QG, GG, a significant decrease of the expression of p-ERK1/2 was obviously observed, especially for GG, indicating QT, HY, IQT, QG, GG at the dosage of 100 μM can effectively inhibit the activation of ERK1/2 and the inhibitive effect was adjacent to that of NADPH oxidase inhibitor DPI, except for QG. Taken consideration together, these data suggested that QT, HY, IQT, QG, GG can prevent high glucose-induced EMT in HK-2 cells via inhibiting NADPH oxidase/ROS/ERK pathway.

4. Discussion

Formation mechanism of adenine-induce rat animal model of CRF (Yokozawa method) is the principle of kidney damage with nephrotoxicity drugs, causing nitrogen qualitative hematic disease, toxin accumulation, electrolyte and amino acid metabolism disorder and resulting in kidney failure eventually. This study show that adenine-induced rat animal model of CRF was characterized by renal tubular injury, renal interstitial fibrosis, glomerular atrophy in part by HE staining, especially renal tubular injury, which was consistent with the reference (Liu et al., 2011). What's more, the previous study exhibited the protective effect of *Abelmosehus Manihot* on rats renal tubular with Nephrotic Syndrome (Yin et al., 2000), but the action mechanism wasn't fully elucidated. Hence, in this study, HK-2 cells were used as a model of proximal tubular epithelium in vitro experiments.

High glucose, the unique biochemical characteristic of diabetic nephropathy, is closely related to functional and structural damages in renal proximal tubular cells (Sun et al., 2008, 2011). Reports demonstrated that high glucose can induce EMT in HK-2 cells and resulted in tubulointerstitial fibrosis (Wei et al., 2013; Wang et al., 2014). Studies indicated that intracellular ROS production is closely associated with tubular EMT (Rhyu et al., 2005; Zhang et al., 2007). Intracellular ROS production is mediated by multiple pathways and mainly derived from NADPH oxidase (Li et al., 2003). Two membrane-bound subunits (p22phox and gp91phox, also called NOX2) is the member of NADPH oxidase (Kleniewska et al., 2012). NOX2 homologs, such as NOX1, NOX3, NOX4, NOX5, Duox1, Duox2 were displayed in non-phagocytic cells (Dworakowski et al., 2008). NOX1, NOX2 and NOX4 were detected in renal cells, and NOX4 was the most abundant (Sedeek et al., 2010). EMT was also involved in the activation of ERK1/2 signaling pathway, which is probably mediated by the transcription factor Snail and by down-regulating of E-cadherin (Nagarajan et al., 2012).

5. Conclusions

In this article, we investigated the renal protective effect and action mechanism of Huangkui capsule and its main five flavonoids. The results stated that Huangkui capsule can alleviate tubulointerstitial fibrosis and regulate the expressions of NOX1, NOX2, NOX4, α-SMA, p-ERK. Flavonoids compounds in Huangkui capsule exerted its potential benefits in preventing high glucose-induced EMT in HK-2 cells by inhibiting ROS production via down-regulation of NADPH oxidase subunits NOX1, NOX2 and NOX4 and ERK1/2 signaling. The data in this paper implied that Huangkui capsule and its active components protect tubulointerstitial fibrosis of CRF involvement in the action mechanism of inhibiting NADPH oxidase/ROS/ERK pathway. These would be important references for clinical treatment of the kidney disease patients and action mechanisms.

Conflicts of Interest

All authors declare there is no conflict of interest.

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